MSG No. 172.128 P_c6_4 [Type IV, hexagonal]

Table 1: Wyckoff site: `6a`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4]`
2	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[2,5]`
3	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[3,6]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,10]`
5	$\left[0,\ 0,\ z+\frac{1}{6}\right]$	`[8,11]`
6	$\left[0,\ 0,\ z+\frac{5}{6}\right]$	`[9,12]`

Table 2: Wyckoff site: `6b`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[2,5]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{3}\right]$	`[3,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,10]`
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{6}\right]$	`[8,11]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{5}{6}\right]$	`[9,12]`

Table 3: Wyckoff site: `12c`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x-y,\ x,\ z+\frac{2}{3}\right]$	`[2]`
3	$\left[-y,\ x-y,\ z+\frac{1}{3}\right]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$\left[-x+y,\ -x,\ z+\frac{2}{3}\right]$	`[5]`
6	$\left[y,\ -x+y,\ z+\frac{1}{3}\right]$	`[6]`
7	$\left[x,\ y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x-y,\ x,\ z+\frac{1}{6}\right]$	`[8]`
9	$\left[-y,\ x-y,\ z+\frac{5}{6}\right]$	`[9]`
10	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[-x+y,\ -x,\ z+\frac{1}{6}\right]$	`[11]`
12	$\left[y,\ -x+y,\ z+\frac{5}{6}\right]$	`[12]`